

ORIGINAL ARTICLE

Hypertriglyceridaemia-associated acute pancreatitis: diagnosis and impact on severity

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Abstract

Background: The level of hypertriglyceridaemia (HTG) at which the risk of acute pancreatitis (AP) increases and the impact of HTG on AP attributable to other aetiologies remains unclear.

Methods: We compared clinical outcomes of patients admitted within 48 h of the onset of abdominal pain from a first episode of AP and admission serum triglyceride levels of either <5.65 mmol/l (<500 mg/dl) or ≥5.65 to <11.3 mmol/l (moderate HTG) or ≥11.3 mmol/l (≥1000 mg/dl, severe HTG).

Results: Among a cohort of 1,233 patients with AP there were significant progressive increases in all major deleterious clinical outcomes including mortality (all $P_{trend} < 0.05$) that were directly dependent on admission triglyceride levels. Outcomes were improved by earlier presentation (<24 h compared to 24–48 h from abdominal pain onset). Patients with severe HTG and a concomitant aetiology ($n = 68$) had significantly more persistent organ failure, pancreatic necrosis and longer hospital stays ($P < 0.05$) than those with severe HTG alone ($n = 206$).

Conclusions: There appears to be an association between HTG grade and the severity of AP. Severe HTG significantly increased the severity of AP, over AP attributable to other aetiologies. Moderate as well as severe HTG can be used as a criterion for the diagnosis of HTG-associated AP.

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Introduction

An important aetiology for acute pancreatitis (AP) is hypertriglyceridaemia (HTG),¹ which may be the sole cause or a co-factor present concomitantly with other causes of AP and/or resulting from other contributors to AP severity that include alcohol abuse, smoking, obesity, metabolic syndrome, type 2 diabetes mellitus and cardiovascular disease.^{2,3} Pooled results from 34 studies found the overall incidence of HTG-associated AP was 9%,⁴ indicating that it is the third most common AP aetiology after gallstones and alcohol.

Most existing clinical guidelines for AP lack a minimum diagnostic serum triglyceride value for HTG-associated AP.^{5–10} The American Practice Guidelines and Endocrine Society recommend ≥11.3 mmol/l (1,000 mg/dl, severe HTG^{11,12}), even though lower levels may cause HTG-associated AP.^{13–15} The Chinese Practice Guidelines recommend ≥11.3 mmol/l or the presence of milky serum.¹⁶ Other studies suggest either ≥11.3 mmol/l or ≥5.65 mmol/l accompanied by milky serum^{17–21} or a previous history of HTG.^{22,23} Systematic review of 38 studies found 15–20% of 1,979 patients with severe HTG developed HTG-associated AP²⁴; in view of this high order of risk it would seem likely that HTG-associated AP can develop below this level.

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Data suggest that HTG-associated AP results in worse clinical outcomes than AP associated with other aetiologies. A recent systematic review and meta-analysis of studies from China, Japan, Mexico, UK and USA comparing the outcome of AP in 1,564 patients with HTG-associated AP and 5,721 patients with other aetiologies found a significantly increased odds ratios for persistent systemic inflammatory response syndrome (SIRS), persistent organ failure and mortality in patients with HTG-associated AP.¹³ Seven of the 14 studies in this meta-analysis used 5.65 mmol/l (500 mg/dl, moderate HTG up to 1000 mg/dl) as the cut-off (minimum) value for the diagnosis of HTG-associated AP, as in a further study of 121 patients with the same findings,¹⁴ but all other studies in this meta-analysis used values < 5.65 mmol/l. The authors concluded that further well-designed and large-scale studies were required to confirm these findings (the included studies each had between 14 and 268 patients with HTG-associated AP,¹³ nor did this meta-analysis address any relationship between the severity of HTG and of AP, or whether HTG contributed to AP associated with a concomitant aetiology. Furthermore, previous studies have not assessed the impact of HTG-associated AP on the frequency of newly diagnosed diabetes mellitus.²⁵

The aim of this present study was to examine the significance of the two recommended diagnostic serum triglyceride thresholds (≥ 5.65 and ≥ 11.3 mmol/l) measured using blood sampled on admission to identify HTG-associated AP in a substantial cohort of 1,233 patients. We sought to determine any relationship between the severity of HTG and that of AP, including on the frequency of newly diagnosed diabetes mellitus, and the impact of HTG in the presence of other concomitant aetiologies for AP.

Methods

Patient selection, clinical setting and ethics

All patients admitted with AP to the Sichuan Provincial Pancreatitis Centre at West China Hospital of Sichuan University from 1st September 2009 to 30th September 2013 were considered for entry into this study. The Institutional Review Board of West China Hospital approved the study protocol, waiving the requirement for informed patient consent because of the retrospective use of data.

Study design

This was a retrospective cohort study of patients with AP either with no or mild HTG (<5.65 mmol/l) or moderate HTG (≥ 5.65 to <11.3 mmol/l) or severe HTG (≥ 11.3 mmol/l), with identification of other aetiologies associated with AP. The reporting of this study followed the STROBE guideline for observational studies.²⁶

Inclusion and exclusion criteria

AP was diagnosed according to the Revised Atlanta Classification definition.⁸ Eligible patients were 18–80 years old and displayed

abdominal pain for <48 h prior to admission regardless of whether they were a primary, secondary or tertiary referral. Patients were excluded if they had any documented/known prior AP episode(s), known history of advanced respiratory, cardiovascular, renal, hepatic, immunological or oncological diseases; and/or trauma, tumour, pregnancy and/or chronic pancreatitis as aetiologies.²⁷

Definition of HTG-associated AP

Fasting serum lipid profiles including triglycerides, total cholesterol, low-density lipoprotein cholesterol and high-density lipoprotein cholesterol were measured immediately after hospital admission in our Clinical Biochemistry Department (almost all patients do not eat after the onset of pain from AP prior to admission). Throughout the period from which the patients were drawn HTG-associated AP (the term used in preference to HTG-AP) was diagnosed when peak triglyceride levels obtained within 24 h of admission were ≥ 5.65 mmol/l.¹⁴ If diagnostic searches made on each patient revealed other aetiologies, a concomitant aetiology was diagnosed e.g. if gallstones or a history of alcohol abuse or other known cause of AP was also present. Patients with peak triglyceride levels <5.65 mmol/l were considered to have developed AP from other causes (non-HTG-associated AP).

Management of AP

The general management of AP patients followed International²⁸ and Chinese guidelines.^{16,29} In mild biliary AP, cholecystectomy was advised in most cases during the index admission, while in severe cases it was postponed until local inflammation controlled and fluid collections resolved or encapsulated. Endoscopic retrograde cholangiopancreatography and endoscopic sphincterotomy were performed within 24 h of admission for patients with cholangitis or obstructive biliary AP. Patients were transferred to the High Dependency Unit (HDU) within our Department if non-invasive ventilation and/or vasoactive agents were required. Patients were referred to the Intensive Care Unit (ICU) if invasive ventilation, inotropes and/or renal replacement therapy were indicated. HTG was managed by a combination of lifestyle recommendations and lipid lowering drugs (principally statins and fibrates^{1,12}), plasmapheresis, heparin or insulin were not used.

Study variables and outcome measures

The definition of study variables and outcome measures are shown in *Supplementary Methods*. Clinical data were extracted from the electronic Health Information System of our hospital using a predefined, standardised proforma. Quality control, quality assurance and standard operating procedures were applied to all stages of data collection to ensure accurate compilation. Data included time of symptom onset to hospital admission, referral status, age, sex, aetiology, alcohol intake, comorbidities, Charlson comorbidity index, biochemical parameters, imaging, surgical interventions, microbiology,

treatments and clinical outcomes. Routine clinical severity scores including SIRS and Sequential Organ Failure Assessment (SOFA) were calculated from all data in the first 24 h and peak values were used. The primary outcome was persistent organ failure. Other outcome measures included multiple organ dysfunction syndrome, HDU/ICU admission, pancreatic necrosis, acute peripancreatic fluid collection, necrosectomy,

newly diagnosed diabetes mellitus, mortality and length of hospital stay.

Statistical analysis

Statistical analyses were performed using SPSS® 22.0 (IBM, Armonk, New York, USA) and Stata 14 (StataCorp, College Station, USA). Categorical data are presented as number and per

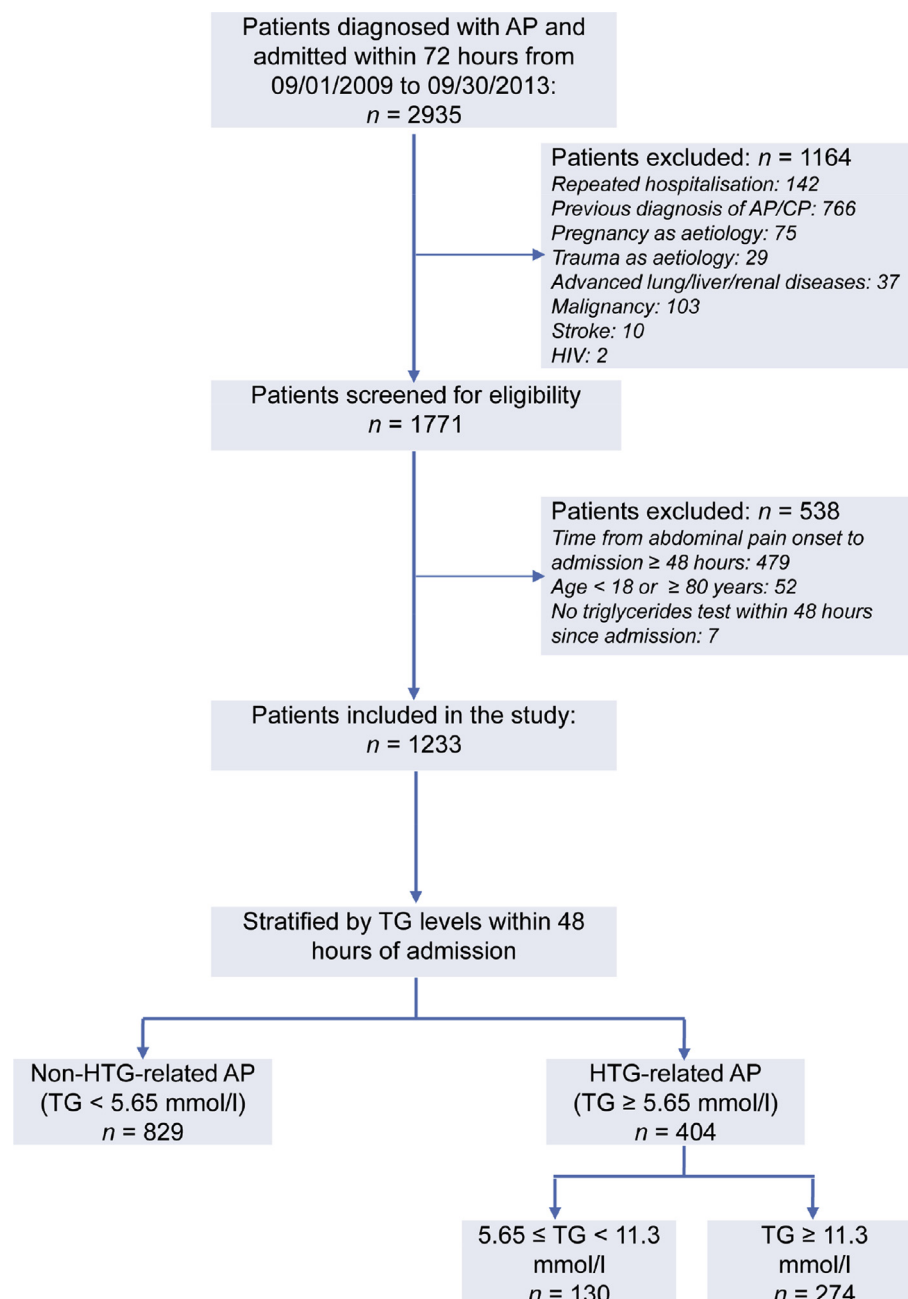


Figure 1 Flow chart showing patient selection process. AP, acute pancreatitis; CP, chronic pancreatitis; HIV, human immunodeficiency virus; TG, triglyceride; HTG, hypertriglyceridaemia

cent and were compared by Chi-square test (or Fisher's test; 2 groups) or proportional trend test (ordered multiple groups). Continuous data are expressed as median with interquartile

range and were compared by Mann–Whitney *U* test (2 groups), Kruskal–Wallis test (3 groups), or Cuzick's trend test (ordered multiple groups). Multivariate logistic regression analysis was

Table 1 Baseline characteristics of AP patients stratified by diagnostic triglyceride levels

Parameters	I: TG < 5.65 mmol/l (n = 829)	II: 5.65 ≤ TG < 11.3 mmol/l (n = 130)	III: TG ≥ 11.3 mmol/l (n = 274)	P Value†
Demographics				
Age (years)*	51 (40.5–63)	41.5 (35–47)	41 (36–46)	<0.001§
Sex (male)	444 (53.6)	102 (78.5)	196 (71.5)	<0.001§
Pre-existing comorbidities				
DM	65 (7.8)	14 (10.8)	52 (19)	<0.001‡
Fatty liver	304 (36.7)	95 (73.1)	198 (72.3)	<0.001§
Charlson comorbidity index*	1 (0–1)	1 (1–1)	1 (1–1)	<0.001§
Aetiologies				
Biliary	529 (63.8)	17 (13.1)	46 (16.8)	<0.001§
Alcoholic	44 (5.3)	11 (8.5)	16 (5.8)	0.356
HTG-associated alone	0 (0)	96 (73.8)	207 (75.5)	<0.001§
Others	256 (30.9)	6 (4.6)	5 (1.8)	<0.001§
Time to admission (h)*	24 (12.5–30)	25 (17.75–33.5)	24 (13–29)	0.005//
Lipid lowering agents	40 (4.8)	30 (16)	82 (29.9)	<0.001§
Lipid profile*				
TG (mmol/l)	1.21 (0.79–2.08)	8.11 (6.53–9.68)	18.52 (15.50–20.16)	<0.001¶
Cholesterol (mmol/l)	3.95 (3.27–4.62)	5.61 (4.62–6.67)	9.64 (7.42–12.71)	<0.001¶
LDL-C (mmol/l)	2 (1.4–2.6)	1.2 (0.7–2)	0.6 (0.3–1.3)	<0.001¶
HDL-C (mmol/l)	1.1 (0.8–1.4)	0.8 (0.7–1.1)	0.9 (0.6–1.1)	<0.001§
Full blood count*				
Haemoglobin (g/l)	144 (130–158)	160 (143–174)	161 (145–172)	<0.001§
Haematocrit (%)	0.43 (0.40–0.46)	0.46 (0.42–0.48)	0.44 (0.40–0.47)	<0.001§
Platelets (× 10 ⁹ /l)	139 (103–175)	135 (102–165)	131 (93–184)	0.194
WBC count (× 10 ⁹ /l)	13.3 (10.1–16.6)	14.0 (11.0–17.6)	14.4 (11.6–17.8)	<0.001§
Neutrophils (× 10 ⁹ /l)	11.6 (8.5–14.8)	12.3 (9.4–15.8)	12.6 (9.7–15.5)	0.001§
Liver and kidney function test*				
Glucose (mmol/l)	8.1 (6.4–11)	9.6 (7–14.4)	12.1 (8.2–17)	<0.001¶
Ca ²⁺ (mmol/l)	2.1 (1.9–2.2)	2.0 (1.7–2.2)	1.9 (1.6–2.1)	<0.001§
Albumin (g/l)	40.8 (36.6–44.0)	40.8 (35.4–43.8)	40 (35.8–42.9)	0.034**
Globulin (g/l)	24.4 (20.8–28.4)	24.9 (22.1–29.7)	25 (21.3–28.9)	0.314
Urea (mmol/l)	5.3 (3.9–7.0)	4.7 (3.5–7.0)	4.7 (3.3–6.1)	<0.001§
Creatinine (μmol/l)	74 (63–89)	80 (65–99)	74 (61–93)	0.092//
ALT (U/l)	52 (24–157)	30 (21–49)	36 (25–57)	<0.001¶
AST (U/l)	50 (26–154)	36 (25–53)	45 (31–69)	<0.001¶
LDH (IU/l)	234 (178–347)	314 (205–494)	350 (244–587)	<0.001¶
Clinical severity scores*				
SIRS	2 (1–2)	2 (2–3)	2 (2–3)	<0.001§
SOFA	0 (0–2)	1 (0–2)	1 (0–2)	0.003§

Values in parentheses are percentages unless indicated otherwise. AP, acute pancreatitis; TG, triglycerides; DM, diabetes mellitus; LDL-C, low-density lipoprotein C; HDL-C, high-density lipoprotein C; WBC, white blood cells; ALT, alanine aminotransferase; AST, aspartate aminotransferase; LDH, lactate dehydrogenase; SIRS, Systemic Inflammatory Response Syndrome; SOFA, Sequential Organ Failure Assessment. *Values are median (interquartile range). † χ^2 or Fisher's exact test for categorical data and Kruskal–Wallis test for continuous data. ‡ $P < 0.050$, III versus II or I; § $P < 0.050$, I versus II or III; ¶ $P < 0.050$, between any two groups; // $P < 0.050$, I versus II; ** $P < 0.050$, I versus III.

Table 2 Clinical outcomes of patients in HTG-associated AP according to different time to admission

Parameters	Time to admission <24 h (n = 177)	Time to admission 24–48 h (n = 227)	P [†]	Estimate ^{‡§}	Adjusted P
Persistent organ failure	54 (30.5)	97 (42.7)	0.012	1.55 (1.00, 2.40)	0.051
MODS	7 (4)	31 (13.7)	0.001	3.80 (1.59, 9.08)	0.003
HDU/ICU admission	19 (10.7)	34 (15)	0.210	1.17 (0.72, 1.88)	0.531
Pancreatic necrosis	37 (20.9)	63 (27.8)	0.113	1.25 (0.77, 2.02)	0.369
Newly diagnosed DM	23 (13)	39 (17.2)	0.247	1.18 (0.66, 2.11)	0.576
Mortality	5 (2.8)	25 (11)	0.002	4.62 (1.66, 12.83)	0.003
Length of hospital stay (d) [*]	11 (8–18)	13 (10–19)	0.026	0.84 (0.68, 1.04) [¶]	0.113
	POF (n = 54)	POF (n = 97)			
Mortality of persistent organ failure	5 (9.3)	25 (25.8)	0.015	3.85 (1.30, 11.39)	0.015

Values in parentheses are percentages unless indicated otherwise. HTG-associated AP, hypertriglyceridaemia-associated acute pancreatitis; MODS, multiple organ dysfunction syndrome; HDU, Highly Dependency Unit; ICU, Intensive Care Unit; DM, diabetes mellitus. *Values are median (interquartile range). [†] χ^2 or Fisher's exact test for categorical data and Mann-Whitney *U* test for continuous data. [‡]Values in parentheses are 95% confidence interval. [§]Multivariate logistic regression model is used with odds ratio reported unless indicated otherwise and [¶]Cox proportional hazards model with hazard ratio reported after adjusting for age, sex, referral, comorbidities, and time from onset of abdominal pain to admission (< 24 h versus 24–48 h)

used to report categorical outcome measures (odds ratios, OR), while Cox proportional hazards analysis with confidence interval (CI) was used to report predicted ICU and hospital discharge (hazard ratios, HR; days for deceased patients were considered as truncated data), respectively. A two-sided $P < 0.05$ was considered statistically significant.

Results

The patient selection process, exclusions and stratification according to fasting serum triglyceride levels are shown in Fig. 1. A total of 1,233 patients had a diagnosis of AP fulfilling the specified criteria out of 2935 patients identified as having been admitted to our centre with AP during the study period. The selected patients comprised 829 patients with normal triglyceride levels or mild HTG, 130 patients with moderate HTG and 274 with severe HTG.

Baseline characteristics of AP patients stratified by diagnostic triglyceride levels

The baseline characteristics for the included patients are shown in Table 1. Patients with severe or moderate HTG were younger than those with no or mild HTG (median age 41 or 41.5 versus 51 years, both $P < 0.050$), included more males (71.5% or 78.5% versus 53.6%, both $P < 0.050$), had more pre-existing fatty liver disease (72.3% or 73.1% versus 36.7%, both $P < 0.050$; reflected in a higher Charlson comorbidity index; both $P < 0.050$), included far fewer with co-existing gallstones (16.8% or 13.1% versus 63.8%, both $P < 0.050$), and were far more frequently in prior receipt of lipid lowering drugs (29.9% or 16% versus 4.8%, both $P < 0.050$). Patients with severe HTG had significantly higher proportion of pre-existing diabetes mellitus than those

with moderate and no or mild HTG (19% versus 10.8% or 7.8%, both $P < 0.050$).

On admission patients with severe HTG, moderate HTG and no or mild had significantly downgraded triglyceride levels (median 18.52 versus 8.11 versus 1.21 mmol/l, all $P < 0.050$) and cholesterol (9.64 versus 5.61 versus 3.95 mmol/l, all $P < 0.050$), while lower low-density lipoprotein cholesterol was inversely increased (0.6 versus 1.2 versus 2.0 mmol/l, all $P < 0.050$). Patient with severe and moderate HTG had significantly lower levels of high-density lipoprotein cholesterol (0.9 or 0.8 versus 1.1 mmol/l, both $P < 0.050$).

Patients with severe and moderate HTG had significantly more deranged biochemical markers with elevated SIRS and SOFA than those with no or mild HTG (both $P < 0.050$).

Time to admission has significant impact on clinical outcomes of HTG-associated AP

The effects of time from the onset of abdominal pain to admission (early: < 24 h versus late: 24–48 h) on study variables and clinical outcomes of HTG-associated AP are summarised (Table S1 and Table 2). In HTG-associated AP, patients in the late admission group had higher proportion of biliary aetiology (19.4% versus 10.7%, $P = 0.017$), lower median triglyceride levels (14.96 versus 16.18 mmol/l, $P = 0.018$) but were associated with higher glucose levels (12.3 versus 10.0 mmol/l, $P < 0.001$) compared to the early admission group (all $P < 0.05$; Table S1). Furthermore, the late admission group had higher incidence of persistent organ failure (42.7% versus 30.5%; OR 1.55, 95% CI 1.00 to 2.40, $P = 0.051$), multiple organ dysfunction syndrome (13.7% versus 4.0%; OR 3.80 [1.59 to 9.08]), overall mortality (11.0% versus 2.8%; OR 4.62 [1.66 to 12.83]) and mortality in persistent organ failure (25.8% versus 9.3%; OR 3.85 [1.30 to

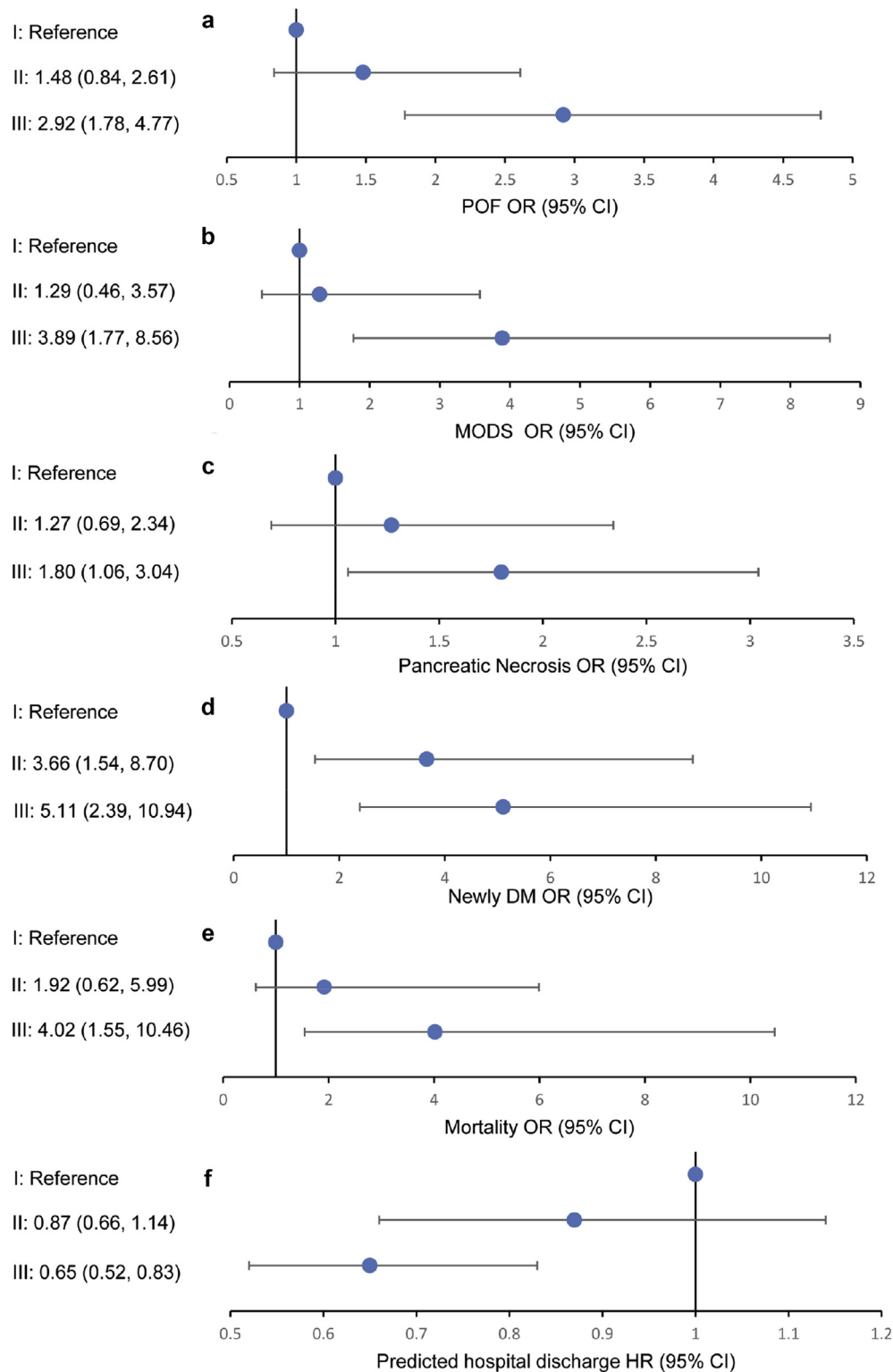


Figure 2 Trend analysis in patients with different triglyceride cut-off values: (a) persistent organ failure (POF); (b) multiple organ dysfunction syndrome (MODS); (c) pancreatic necrosis; (d) newly diagnosed diabetes mellitus (DM); (e) mortality and (f) predicted hospital discharge hazard ratio. I: triglyceride levels <5.65 mmol/l; II: ≥ 5.65 mmol/l to <11.3 mmol/l; III: TG ≥ 11.3 mmol/l. OR, odds ratio; HR, hazard ratio

Table 3 Clinical outcomes of HTG-associated AP with or without other aetiology

A					
Parameters	5.65 ≤ TG < 11.3 mmol/l with HTG alone (n = 96)	5.65 ≤ TG < 11.3 mmol/l with other aetiology (n = 34)	P [†]	Estimate ^{‡§}	Adjusted P
Persistent organ failure	29 (30.2)	10 (29.4)	1.000	0.91 (0.37, 2.26)	0.842
MODS	5 (5.2)	2 (5.9)	1.000	0.88 (0.14, 5.36)	0.886
HDU/ICU admission	31 (32.3)	10 (29.4)	0.832	0.83 (0.33, 2.07)	0.692
Pancreatic necrosis	23 (24)	6 (17.6)	0.632	0.80 (0.28, 2.29)	0.677
Newly diagnosed DM	10 (10.4)	7 (20.6)	0.146	2.78 (0.79, 9.80)	0.112
Mortality	5 (5.2)	2 (5.9)	1.000	0.79 (0.13, 4.67)	0.793
Length of hospital stay (d) [*]	11 (9–16)	13 (7.8–18.3)	0.951	1.16 (0.75, 1.79) [¶]	0.499
B					
Parameters	TG ≥ 11.3 mmol/l with HTG alone (n = 206)	TG ≥ 11.3 mmol/l with other aetiology (n = 68)	P [†]	Estimate ^{‡§}	Adjusted P
Persistent organ failure	74 (35.9)	38 (55.9)	0.004	1.88 (1.34, 3.39)	0.037
MODS	20 (9.7)	11 (16.2)	0.184	1.25 (0.53, 2.97)	0.607
HDU/ICU admission	79 (38.3)	36 (52.9)	0.047	1.47 (0.81, 2.67)	0.203
Pancreatic necrosis	46 (22.3)	25 (36.8)	0.025	1.71 (0.82, 3.16)	0.090
Newly diagnosed DM	36 (17.5)	9 (13.2)	0.457	0.73 (0.31, 1.71)	0.474
Mortality	16 (7.8)	7 (10.3)	0.614	0.81 (0.28, 2.31)	0.686
Length of hospital stay (d) [*]	13 (8–18.3)	14.5 (11–24.8)	0.016	0.82 (0.60, 1.11) [¶]	0.196

Values in parentheses are percentages unless indicated otherwise. HTG-associated AP, hypertriglyceridaemia-associated acute pancreatitis; MODS, multiple organ dysfunction syndrome; HDU, Highly Dependency Unit; ICU, Intensive Care Unit; DM, diabetes mellitus. *Values are median (interquartile range). † χ^2 or Fisher's exact test for categorical data and Mann-Whitney *U* test for continuous data. ‡Values in parentheses are 95% confidence interval. §Multivariate logistic regression model is used with odds ratio reported unless indicated otherwise and ¶Cox proportional hazards model with hazard ratio reported after adjusting for age, sex, referral, comorbidities, and time from onset of abdominal pain to admission (< 24 h versus 24–48 h).

11.39]) compared with early admission group (all remaining $P < 0.05$).

Clinical outcomes of AP patients stratified by diagnostic triglyceride levels

Subsequent comparisons of clinical outcomes included adjusting baseline parameters of time from the onset of pain to admission (<24 h and 24–48 h) as well as age, gender, route of referral, other aetiological factors and co-morbidities.

The results of trend analyses of clinical outcomes for these three groups of patients are depicted in Fig. 2A–F. There was a significant, progressive increase in the incidence of persistent organ failure, multiple organ dysfunction syndrome, pancreatic necrosis, newly diagnosed diabetes mellitus and mortality as the severity of HTG increased (all $P_{trend} < 0.05$). The predicted hospital discharge significantly and progressively decreased with an increasing severity of HTG ($P_{trend} < 0.001$), indicating progressively longer hospital stays with increasing HTG on admission for AP. There were no significant differences for acute peripancreatic fluid collection (all $P_{trend} > 0.1$).

Direct comparison between patients with no or mild HTG and moderate HTG showed a trend towards an increased incidence of

persistent organ failure, multiple organ dysfunction syndrome, pancreatic necrosis and mortality with moderate HTG (OR 1.27–1.92, all $P > 0.05$), with a statistically significant increase of newly diagnosed diabetes mellitus with moderate HTG (OR 4.37 [7.72 to 11.10], $P = 0.002$); there was also a trend towards less predicted hospital discharge with moderate HTG (HR 0.87 [0.66–1.14], $P > 0.05$). Comparison of patients with no or mild HTG and severe HTG showed that severe HTG resulted in a significantly increased incidence of persistent organ failure (OR 2.92 [1.78 to 4.77]), multiple organ dysfunction syndrome (OR 3.89 [1.77 to 8.56]), pancreatic necrosis (OR 1.80 [1.06 to 3.04]), newly diagnosed diabetes mellitus (OR 5.11 [2.39 to 10.94]) and mortality (OR 4.02 [1.55 to 10.46]) (all $P < 0.05$), as well as a significantly reduced predicted hospital discharge (HR 0.65 [0.52–0.83], indicating longer hospital stays).

HTG-associated AP with or without concomitant aetiologies

Direct comparison of the clinical outcomes of HTG-associated AP without any other identified aetiological factor *versus* HTG-associated AP in the presence of another identified aetiology showed a significant increase in persistent organ failure and

length of hospital stay in the latter group by univariate analysis, although significance disappeared after adjusting for age, gender, route of referral, aetiologies, co-morbidities and time from onset of abdominal pain to admission by multivariate analysis (Table S2). Further analyses however showed that the significant increase in persistent organ failure persisted after these adjustments in multivariate analysis when patients with severe HTG and no other identified aetiological factor were compared with severe HTG-associated AP in the presence of another identified aetiology (OR 1.88 [1.34 to 3.39], $P = 0.037$, Table 3).

Discussion

This study describes the largest single centre cohort of patients with HTG-associated AP reported to date, which has enabled us to make novel findings that bear upon our understanding and management of HTG-associated AP. We found patients admitted with AP and moderate HTG (serum triglyceride level ≥ 5.65 to <11.3 mmol/l) or severe HTG (>11.3 mmol/l) to have a number of presenting features that were significantly different from patients with AP and no or mild HTG (<5.65 mmol/l). In comparison to patients with no or mild HTG, those with moderate or severe HTG were significantly younger, significantly more often male and had significantly more frequent pre-existing diabetes mellitus, fatty liver, deranged lipid profiles and worse disease severity predictors at presentation. Despite there was only a tendency of increased AP severity with moderate HTG, the severity became markedly and significantly increased in severe HTG. These features indicate HTG-associated AP is different from AP attributable to other aetiologies, and support our use of a serum triglyceride level of 5.65 mmol/l rather than 11.3 mmol/l as the diagnostic cut-off for this aetiology, whereas mild HTG may be an epiphenomenon.^{30,31} If mild HTG was an aetiological factor¹⁵ similar to but less marked than moderate or severe HTG in any patients whom we allocated to the non-HTG-associated AP group, the number affected was insufficient to remove these differences. Thus while a number of authorities have stated that confirmation of severe HTG is necessary to make the diagnosis of HTG-associated AP,^{11,12} our data indicate that the presence of moderate HTG (≥ 5.65 mmol/l) is sufficient for this purpose to alert specific management.

Consistent with studies that have suggested a direct relationship between the severity of HTG and of AP,^{13,14,23} we found all major clinical outcomes were significantly and progressively worse for patients with increasing triglyceride levels, as shown by comparison of persistent organ failure, multiple organ dysfunction syndrome, pancreatic necrosis, new diagnoses of diabetes mellitus and most notably mortality for those with no or mild, moderate and severe HTG. We also found patients with HTG-associated AP who were admitted within 24 h after the onset of pain had significantly better clinical outcomes than patients admitted from 24 to 48 h

after the onset of pain, underlining the importance of prompt diagnosis and treatment to improve clinical outcomes from HTG-associated AP.³² Of particular note, when HTG-associated AP was accompanied by another aetiology, the frequency of persistent organ failure was significantly higher than in HTG-associated AP without another aetiology. While the presence of gallstones or alcohol abuse increases the likelihood of AP in patients with HTG over the risk attendant upon HTG alone,³³ our study provides new data consistent with HTG and another aetiology together contributing to the pathogenesis and/or severity of AP. Most often AP is assumed to result from a single aetiology,^{5–11} but preclinical data have demonstrated that two co-factors may operate together to induce pancreatic acinar cell damage or increase the severity of experimental AP, e.g. fatty acid ethyl ester together with ethanol or a secretagogue at concentrations that individually have modest effects.^{34,35} In any case if two aetiologies are identified in any patient, both should be addressed to reduce the likelihood of recurrent AP.

Here we measured fasting serum triglycerides on admission, providing the most direct evidence of HTG in the aetiology of AP. This approach may account for the relatively high proportion of patients diagnosed with HTG-associated AP (32.8% with 22.2% ≥ 11.3 mmol/l) compared to some other studies (8–26%) in China.⁴ If triglycerides are measured at a later time point, case ascertainment is likely to decrease; in one study the sensitivity and specificity of the detection of HTG-associated AP was estimated to have decreased by one half when measurement was undertaken on day 2 after admission.¹⁸ Guidelines are largely silent on the relative benefits of alternative sampling protocols to diagnose HTG-associated AP.^{5–10,16} Triglyceride elevations may not be as marked during or after recovery e.g. as a result of modified and/or reduced oral intake. While fasting triglyceride levels remain the most widely recommended test for HTG,¹² non-fasting triglyceride levels may be equally informative, as these are subject to less variability than previously thought, can be combined with direct measurement of low density lipoprotein cholesterol (rather than by the Friedewald equation that requires fasting samples) and might better reflect both total blood triglyceride load over any 24 h period and cumulative disease risk.^{36–38} Fat tolerance testing may also be used,^{37,39} but has not been widely applied in the assessment of HTG-associated AP.³¹ Further studies are warranted to capture the changes of triglyceride levels during the clinical course of HTG-associated AP.

Fewer than one third of our patients diagnosed with HTG-associated AP were taking lipid lowering agents prior to their admission, because HTG had been unrecognised or previously considered less severe or treatment had been unavailable or declined e.g. because of cost. Optimal treatment of HTG-associated AP has yet to be defined, although in general reduction of serum triglycerides to below 5.65 mmol/l is advised,¹⁶ since this

substantially limits the risk of recurrent AP as well as other complications of HTG.⁴⁰ Our study highlights the need to determine the most effective strategies for early and long-term management of HTG-associated AP, including the indications for immediate measures such as plasmapheresis or insulin, for lipid lowering therapy after hospital discharge such as statins⁴¹ and omega-3 carboxylic acids,⁴² determination of the cause of HTG and the role of genomic analyses.⁴³ While many guidelines recommend measures to reduce the risk of AP in patients with severe HTG,^{13,14,23} our data suggest attention should also be paid to patients with moderate HTG as there is still a significant risk of AP if serum triglyceride levels are ≥ 5.65 mmol/l but below 11.3 mmol/l.

As our study was retrospective, there is a risk of bias in our findings. Our hospital is the largest centre managing AP in China with a large tertiary referral practice. To reduce bias arising from tertiary referrals, we restricted this study to patients with a first episode of AP and onset of abdominal pain within 48 h of admission. Even then our cohort was of sufficient size to show significantly better clinical outcomes for those patients admitted <24 h compared with 24–48 h of the onset of abdominal pain. While we have determined rates of newly diagnosed diabetes mellitus, we have not addressed recurrent and/or chronic HTG-associated pancreatitis, which would comprise a separate study. Again, this highlights the need for much more research in HTG-associated AP, including pathogenesis, natural history and management to improve outcomes from one of the most severe forms of AP.

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Conflicts of interest

RS is a consultant for AbbVie, CalciMedica, Cypralis, GlaxoSmithKline, Lilly and Novartis; RS has received research funding from Cypralis, GSK and Merck; all resources and/or funds for the above have been gifted/paid to the University of Liverpool and/or Royal Liverpool and Broadgreen University Hospitals NHS Trust.

References

1. Tsuang W, Navaneethan U, Ruiz L, Palascak JB, Gelrud A. (2009) Hypertriglyceridemic pancreatitis: presentation and management. *Am J Gastroenterol* 104:984–991.
2. Brahm A, Hegele RA. (2013) Hypertriglyceridemia. *Nutrients* 5: 981–1001.
3. Tada H, Kawashiri MA, Nakahashi T, Yagi K, Chujo D, Ohbatake A *et al.* (2015) Clinical characteristics of Japanese patients with severe hypertriglyceridemia. *J Clin Lipidol* 9:519–524.
4. Carr RA, Rejowski BJ, Cote GA, Pitt HA, Zyromski NJ. (2016) Systematic review of hypertriglyceridemia-induced acute pancreatitis: a more virulent etiology? *Pancreatol* 16:469–476.
5. Working party of the British society of gastroenterology, Association of surgeons of Great Britain and Ireland, Pancreatic society of Great Britain and Ireland, Association of upper GI surgeons of Great Britain and Ireland. (2005) UK guidelines for the management of acute pancreatitis. *Gut* 54(Suppl 3). iii1–9.
6. Koizumi M, Takada T, Kawarada Y, Hirata K, Mayumi T, Yoshida M *et al.* (2006) JPN Guidelines for the management of acute pancreatitis: diagnostic criteria for acute pancreatitis. *J Hepatobiliary Pancreat Surg* 13:25–32.
7. Italian Association for the Study of the Pancreas, Pezzilli R, Zerbi A, Campa D, Capurso G, Golfieri R *et al.* (2015) Consensus guidelines on severe acute pancreatitis. *Dig Liver Dis* 47:532–543.
8. Banks PA, Bollen TL, Dervenis C, Gooszen HG, Johnson CD, Sarr MG *et al.* (2013) Classification of acute pancreatitis–2012: revision of the Atlanta classification and definitions by international consensus. *Gut* 62: 102–111.
9. Tsiotos. (2013) IAP/APA evidence-based guidelines for the management of acute pancreatitis. *Pancreatol* 13:e1–e15.
10. Greenberg JA, Hsu J, Bawazeer M, Marshall J, Friedrich JO, Nathens A *et al.* (2016) Clinical practice guideline: management of acute pancreatitis. *Can J Surg* 59:128–140.
11. Tenner S, Baillie J, DeWitt J, Vege SS. (2013) American College of Gastroenterology guideline: management of acute pancreatitis. *Am J Gastroenterol* 108:1415–1416.
12. Berglund L, Brunzell JD, Goldberg AC, Goldberg IJ, Sacks F, Murad MH *et al.* (2012) Evaluation and treatment of hypertriglyceridemia: an endocrine society clinical practice guideline. *J Clin Endocrinol Metab* 97:2969–2989.
13. Wang Q, Wang G, Qiu Z, He X, Liu C. (2017) Elevated serum triglycerides in the prognostic assessment of acute pancreatitis: a systematic review and meta-analysis of observational Studies. *Elevated serum triglycerides in the prognostic assessment of acute pancreatitis. J Clin Gastroenterol* 51:586–593.
14. Vipperla K, Somerville C, Furlan A, Koutroumpakis E, Saul M, Chennat J *et al.* (2017) Clinical profile and natural course in a large cohort of patients with hypertriglyceridemia and pancreatitis. *J Clin Gastroenterol* 51:77–85.
15. Pedersen SB, Langsted A, Nordestgaard BG. (2016) Nonfasting mild-to-moderate hypertriglyceridemia and risk of acute pancreatitis. *JAMA Intern Med* 176:1834–1842.
16. Pancreatic Surgery BoS, Chinese Medical Association. (2015) Guidelines for diagnosis and treatment of acute pancreatitis. *Clin J Surg* 53: 50–53.
17. Yin G, Hu G, Cang X, Yu G, Hu Y, Xing M *et al.* (2014) C-reactive protein. *Pancreas* 43:1323–1328.
18. Yin G, Cang X, Yu G, Hu G, Ni J, Xiong J *et al.* (2015) Different clinical presentations of hyperlipidemic acute pancreatitis. *Pancreas* 44: 1105–1110.
19. An F, Zhan Q, Xia M, Jiang L, Lu G, Huang M *et al.* (2014) From moderately severe to severe hypertriglyceridemia induced acute pancreatitis: circulating miRNAs play role as potential biomarkers. *PLoS One* 9, e111058.
20. Li MQ, Shi ZX, Xu JY, Lu B, Li JQ, Xu YJ *et al.* (2014) Hemodiafiltration combined with resin-mediated absorption as a therapy for hyperlipidemic acute pancreatitis. *Cell Biochem Biophys* 69:699–702.

21. Huang YX, Jia L, Jiang SM, Wang SB, Li MX, Yang BH. (2014) Incidence and clinical features of hyperlipidemic acute pancreatitis from Guangdong, China. *Pancreas* 43:548–552.
22. Wu C, Ke L, Tong Z, Li B, Zou L, Li W *et al.* (2014) Hypertriglyceridemia is a risk factor for acute kidney injury in the early phase of acute pancreatitis. *Pancreas* 43:1312–1316.
23. Nawaz H, Koutroumpakis E, Easler J, Slivka A, Whitcomb DC, Singh VP *et al.* (2015) Elevated serum triglycerides are independently associated with persistent organ failure in acute pancreatitis. *Am J Gastroenterol* 110:1497–1503.
24. Adiamah A, Psaltis E, Crook M, Lobo DN. (2017) A systematic review of the epidemiology, pathophysiology and current management of hyperlipidaemic pancreatitis. *Clin Nutr*. pii: S0261-5614(17)31360-31362.
25. Das SLM, Singh PP, Phillips ARJ, Murphy R, Windsor JA, Petrov MS. (2014) Newly diagnosed diabetes mellitus after acute pancreatitis: a systematic review and meta-analysis. *Gut* 63:818–831.
26. von EE, Altman DG, Egger M, Pocock SJ, Gøtzsche PC, Vandenbroucke JP *et al.* (2007) Strengthening the Reporting of observational studies in epidemiology (STROBE) statement: guidelines for reporting observational studies. *BMJ* 335:806–808.
27. Liu T, Huang W, Szatmary P, Abrams ST, Alhamdi Y, Lin Z *et al.* (2017) Accuracy of circulating histones in predicting persistent organ failure and mortality in patients with acute pancreatitis. *Br J Surg* 104: 1215–1225.
28. Banks PA, Freeman ML, Practice Parameters Committee of the American College of Gastroenterology. (2006) Practice guidelines in acute pancreatitis. *Am J Gastroenterol* 101:2379–2400.
29. General Surgery Committee CAotIoT, Western M.. (2007) Practical guidelines for management of severe acute pancreatitis with integrated Chinese traditional and Western medicine (Draft). *Zhongguo Wei Zhong Bing Ji Jiu Yi Xue* 19:448–451.
30. Chen WJ, Sun XF, Zhang RX, Xu MJ, Dou TH, Zhang XB *et al.* (2017) Hypertriglyceridemic acute pancreatitis in emergency department: typical clinical features and genetic variants. *J Dig Dis* 18:359–368.
31. Dominguez-Munoz JE, Junemann F, Malfertheiner P. (1995) Hyperlipidemia in acute pancreatitis. Cause or epiphenomenon? *Int J Pancreatol* 18:101–106.
32. Jin T, Huang W, Yang XN, Xue P, Javed MA, Altai K *et al.* (2014) Validation of the moderate severity category of acute pancreatitis defined by determinant-based classification. *Hepatobiliary Pancreat Dis Int* 13: 323–327.
33. Sutton A, Mohanane D, Morkos M, Basu S, Abegunde AT, Ganesh M *et al.* (2018) Acute pancreatitis in patients with severe hypertriglyceridemia in a multi-ethnic minority population. *Endocr Pract* 24:429–436.
34. Huang W, Booth DM, Cane MC, Chvanov M, Javed MA, Elliott VL *et al.* (2014) Fatty acid ethyl ester synthase inhibition ameliorates ethanol-induced Ca²⁺-dependent mitochondrial dysfunction and acute pancreatitis. *Gut* 63:1313–1324.
35. Mukherjee R, Mareninova OA, Odnokova IV, Huang W, Murphy J, Chvanov M *et al.* (2016) Mechanism of mitochondrial permeability transition pore induction and damage in the pancreas: inhibition prevents acute pancreatitis by protecting production of ATP. *Gut* 65: 1333–1346.
36. Mihas C, Kolovou GD, Mikhailidis DP, Kovar J, Lairon D, Nordestgaard BG *et al.* (2011) Diagnostic value of postprandial triglyceride testing in healthy subjects: a meta-analysis. *Curr Vasc Pharmacol* 9:271–280.
37. Kolovou GD, Mikhailidis DP, Kovar J, Lairon D, Nordestgaard BG, Ooi TC *et al.* (2011) Assessment and clinical relevance of non-fasting and postprandial triglycerides: an expert panel statement. *Curr Vasc Pharmacol* 9:258–270.
38. Berglund L, Brunzell JD, Goldberg AC, Goldberg IJ, Stalenhoef A. (2014) Treatment options for hypertriglyceridemia: from risk reduction to pancreatitis. *Best Pract Res Clin Endocrinol Metabol* 28:423–437.
39. O'Doherty AF, Sathyapalan T, Rigby AS, Ingle L, Carroll S. (2018) The repeatability of the abbreviated (4-h) Oral Fat Tolerance Test and influence of prior acute aerobic exercise. *Eur J Nutr* 57:309–318.
40. Christian JB, Arondekar B, Buysman EK, Johnson SL, Seeger JD, Jacobson TA. (2012) Clinical and economic benefits observed when follow-up triglyceride levels are less than 500 mg/dL in patients with severe hypertriglyceridemia. *J Clin Lipidol* 6:450–461.
41. de-Madaria E. (2017) Statins for the prevention of acute pancreatitis. *Am J Gastroenterol* 112:1765–1767.
42. Stroes ESG, Susekov AV, de Bruin TWA, Kvarnstrom M, Yang H, Davidson MH. (2017) Omega-3 carboxylic acids in patients with severe hypertriglyceridemia: EVOLVE II, a randomized, placebo-controlled trial. *J Clin Lipidol* 12:321–330.
43. Whitcomb DC. (2010) Genetic aspects of pancreatitis. *Annu Rev Med* 61:413–424.

Appendix A. Supplementary data

Supplementary data to this article can be found online at <https://doi.org/10.1016/j.hpb.2019.01.015>.